

ASSIUT INTL, Egypt

WMO: 623930

Lat: 27.047N

Lon: 31.012E

Elev: 772

StdP: 14.29

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 40.6	(c) 42.6	(d) 21.4	(e) 16.5	(f) 61.2	(g) 25.0	(h) 19.6	(i) 61.8	(j) 23.1	(k) 62.2	(l) 21.0	(m) 60.9	(n) 4.9	(o) 330	(p) 0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.4	107.3	84.0	66.4	100.0	83.4	64.6	93.7	83.4	38.8	96.7	37.4	95.2	36.3	94.3	87.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 22.6	(b) 20.4	(c) 18.5	(e) 36.4	(f) 112.9	(g) 2.5	(h) 3.3	(i) 34.6	(j) 115.3	(k) 33.1	(l) 117.2	(m) 31.7	(n) 119.1	(o) 29.9	(p) 121.5		
(a) 22.6	(b) 20.4	(c) 18.5	(e) 32.0	(f) 77.2	(g) 2.5	(h) 3.2	(i) 30.2	(j) 79.5	(k) 28.7	(l) 81.4	(m) 27.3	(n) 83.2	(o) 25.5	(p) 85.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	73.5	55.3	59.2	66.0	74.0	81.7	85.9	87.6	87.4	83.1	76.4	66.7	58.2
	DBStd	12.61	4.38	6.30	7.13	6.78	6.03	4.47	3.34	3.09	3.90	5.16	5.28	4.50
	HDD50	7	5	1	0	0	0	0	0	0	0	0	0	1
	HDD65	822	302	187	70	4	0	0	0	0	0	0	39	220
	CDD50	8593	170	258	497	719	982	1077	1164	1160	994	818	500	254
	CDD65	3935	2	25	102	273	517	627	699	695	544	354	89	8
	CDH74	53860	72	378	1414	3423	7081	9191	10522	10283	6892	3599	870	135
	CDH80	31094	9	136	654	1766	4171	5690	6560	6316	3811	1657	288	36
Wind	WSAvg	9.1	7.9	8.7	9.9	9.9	9.5	10.7	9.3	9.0	10.0	9.1	7.8	8.1
Precipitation	PrecAvg	0.3	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0
	PrecMax	2.4	0.3	0.0	0.4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.0	0.4
	PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.4	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.7	90.5	99.9	104.9	110.8	111.1	108.1	107.7	105.5	101.5	93.3	84.3
		MCWB	57.4	61.1	63.9	65.6	69.8	70.8	71.1	72.9	70.6	68.0	64.4	60.8
	2%	DB	75.3	84.5	93.1	100.3	105.4	106.2	104.5	104.0	101.4	96.4	87.4	77.2
		MCWB	55.7	58.6	61.6	64.7	69.2	69.5	70.6	71.2	70.1	68.2	63.8	58.2
	5%	DB	70.9	78.2	87.7	95.2	100.8	102.5	102.1	101.5	98.3	92.6	82.6	72.0
		MCWB	54.1	56.6	60.3	63.0	67.4	69.2	70.1	70.7	70.1	67.6	62.5	56.4
	10%	DB	66.6	73.1	81.6	90.3	96.9	99.3	99.7	98.9	95.0	88.6	78.8	69.0
		MCWB	52.8	55.0	58.3	61.5	65.9	68.5	69.6	70.0	69.2	66.2	61.1	55.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.7	63.6	66.0	68.3	73.8	75.3	78.4	76.3	75.2	72.6	68.0	63.4
		MCDB	75.6	86.5	91.9	97.8	102.5	101.1	94.9	97.6	96.9	91.9	85.2	79.9
	2%	WB	56.8	60.0	63.0	66.1	71.2	72.8	74.7	74.4	73.3	70.4	65.6	59.8
		MCDB	71.8	79.4	88.3	94.5	99.3	98.9	95.3	96.1	93.7	89.9	81.0	73.3
	5%	WB	54.8	57.7	60.9	64.2	68.8	71.0	72.9	73.1	71.7	68.8	63.7	57.6
		MCDB	68.2	75.4	85.0	91.6	95.8	96.9	94.7	94.9	91.6	88.0	79.4	70.1
	10%	WB	53.3	55.5	58.8	62.4	67.1	69.5	71.4	71.9	70.4	67.3	62.0	55.9
		MCDB	65.9	71.7	79.9	88.0	93.9	94.6	93.5	93.7	90.2	85.7	77.2	67.5
Mean Daily Temperature Range	5% DB	MDBR	23.6	24.8	26.2	27.0	27.0	25.7	24.1	23.7	23.7	23.7	24.0	23.2
		MCDBR	29.5	32.0	34.0	33.5	32.2	30.1	26.9	26.4	27.6	27.7	28.5	27.3
		MCWBR	15.1	15.1	14.5	13.2	12.7	10.5	9.8	9.6	9.8	10.0	12.4	13.9
	5% WB	MCDBR	26.5	29.4	31.4	30.9	29.5	27.0	24.6	24.4	24.3	24.9	25.7	24.8
MCWBR		14.3	14.6	14.0	13.0	12.2	10.6	10.1	9.7	9.9	9.9	12.1	13.3	
Clear-Sky Solar Irradiance	taub	0.384	0.416	0.460	0.542	0.558	0.483	0.473	0.464	0.467	0.469	0.401	0.384	
	taud	2.085	1.982	1.864	1.697	1.687	1.882	1.929	1.958	1.935	1.912	2.065	2.094	
	Ebn at Noon	268	269	264	246	241	259	261	263	258	248	259	261	
	Edh at Noon	44	53	63	77	78	64	61	59	58	56	45	42	
All-Sky Solar Radiation	RadAvg	1376	1675	2061	2333	2496	2633	2560	2399	2151	1782	1464	1282	
	RadStd	36	61	64	50	57	27	26	27	45	29	31	25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	+4.77	N/A	+2.32	+2.05	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)	+0.72	+0.65	N/A	N/A	N/A	+0.83	N/A	N/A	N/A	N/A

Nomenclature: See separate page